Inline vs Display Math

Inline equations are written within single dollar signs or parentheses, such as $x + y = 10$. Display equations are centered on their own lines using brackets or environments:

$$E = mc^2$$

Mathematical Fonts and Symbols

Different typefaces can be applied to mathematical text:

Calligraphic:	$\mathcal{RQSZ}$
Fraktur:	$\mathfrak{RQGZ}$
Blackboard Bold:	$\mathbb{R}, \mathbb{Q}$

Greek letters and logic symbols include $\alpha, \beta, \gamma, \pi, \theta$, along with set operations like $A \cap B$ and $A \cup B$.

Calculus and Core Structures

Fractions, roots, and calculus operators are structured as follows:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \quad \int_0^{\infty} e^{-x}\,dx = 1, \quad \sqrt[3]{y}$$